Jonathan Zhu

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Experience

Software Engineer Intern — Royal Bank of Canada May 2026 - Present
Python, TypeScript, FastAPI, LangChain, LangGraph, Kubernetes *Toronto ON, Canada*

- Selected out of **12,000+** applicants for RBC's Amplify innovation program to deliver a **patent-pending** generative AI solution resolving finance modernization bottlenecks.
- Led the complete **0-to-1 product lifecycle** by gathering cross-functional business requirements, designing cloud-native and scalable system architecture, building core application logic using **Python** and **FastAPI**, and demoing end-to-end product flow to C-suite executives.
- Developed an explainable, transparent **agentic AutoML framework** orchestrated using **LangGraph**, reducing model **iteration time** by **99%** and improving model **accuracy** by **88%** compared to existing options.

Embedded Systems Developer — aUToronto (UofT Autonomous Vehicle Team) Sept 2025 - Present
C++, CMake, Linux, ROS2

- Developed fault-tolerant ROS2 and embedded systems infrastructure that supported aUToronto's **first-place overall** finish among **10+ teams** across North America at the 2026 GM-sponsored SAE AutoDrive Challenge.
- Implemented **ring buffer-based serial communication** in C++ to efficiently handle asynchronous data transfer between server and microcontroller, ensuring consistent data handling during high-frequency data transfer.
- Designed a lightweight **binary protocol** to communicate sensor health messages between server and microcontroller, enabling reliable low-latency status synchronization across embedded vehicle systems.

Backend Engineer Intern — PointClickCare Sep 2025 - Dec 2025
Java, Kotlin, Spring Boot, SQL Server, Kubernetes, Redis, Terraform *Toronto ON, Canada*

- Contributed to **3 core microservices** by optimizing APIs and implementing new features using **Spring Boot**, **Kubernetes** and **Terraform**, improving reliability and scalability.
- Achieved projected annual cloud **cost savings** of **$3,000** from identifying and resolving a critical SQL Server database bottleneck by examining SQL Server Query Execution Plans, reducing **CPU utilization** by **36%**.
- Implemented a **Spring Boot** API retry system using Spring Retry library for a new patient responder service, improving fault tolerance and reducing the occurrence of unresolved **transient errors** by **95%**.

Software Engineer Intern — Telesat (LEO Satellite Connectivity Provider) May 2025 - Aug 2025
C++, gRPC, Python, Selenium, Apache Kafka, GitLab CI/CD *Ottawa ON, Canada*

- Designed and implemented an asynchronous **gRPC** file transfer **microservice** using **C++** to support fast and secure file transfers between management system and embedded devices.
- Wrote custom **Selenium** tooling using **Pytest fixtures/hooks** and test tooling for **Apache Kafka** that will be used by **10+** developers estimated to save ~**780** developer hours a year.

Projects

Banter Mart - API Winner @ LA Hacks React, TypeScript, Supabase, Google Gemini

- Won the Best World API Integration prize among **1,400+** participants at LA Hacks 2026 by integrating the World ID API to a agent-native online marketplace.
- Developed a full-stack commerce app using **React**, **TypeScript**, and **Supabase** that replaces traditional marketplace friction with an agentic negotiation interface, allowing buyers and sellers to align on prices effortlessly.

Technical Skills

Languages: Python, C, C++, Java, Kotlin, Rust, JavaScript, TypeScript, SQL, Protobuf
Frameworks/Libraries: React, FastAPI, Flask, Spring Boot, SQLAlchemy, gRPC, Express.js, Langchain, Pytest, Django
Technologies: Git, Bash, Docker, AWS, Azure, Kubernetes, PostgreSQL, Apache Kafka, Terraform, Redis

Education

University of Toronto 2024-2028
*Honours B.Sc., Computer Science & Statistics — **3.8 GPA***

- $3000 renewable entrance scholarship

Relevant coursework: Intro to Linux programming, Data structures and algorithms